Part III

6 High Order Mass Assignment

- Nearest Grid Point (NGP)

$$W^{(1)}(s) = \begin{cases} 1 & \text{for } |s| < \frac{1}{2} \\ 0 & \text{otherwise} \end{cases} \tag{6}$$

- Cloud-In-Cell (CIC)

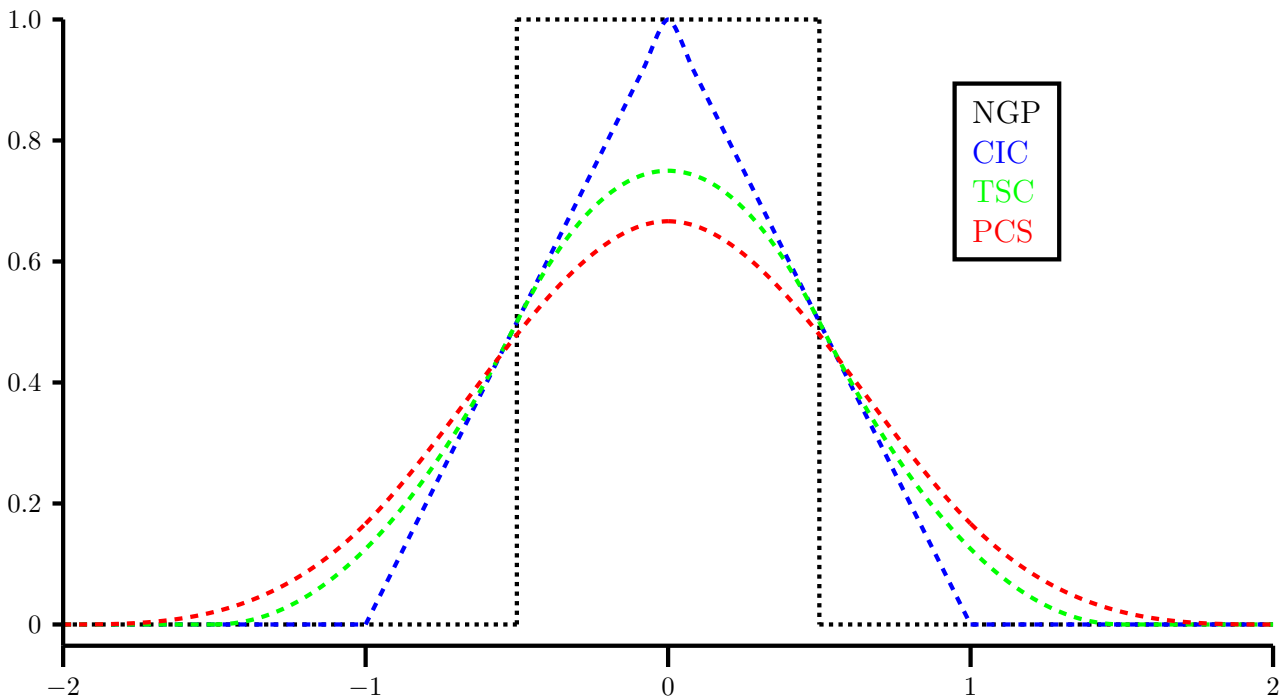
$$W^{(2)}(s) = \begin{cases} 1 - |s| & \text{for } |s| < 1 \\ 0 & \text{otherwise} \end{cases} \tag{7}$$

- Triangular Shaped Cloud (TSC)

$$W^{(3)}(s) = \begin{cases} \frac{3}{4} - s^2 & \text{for } |s| < \frac{1}{2} \\ \frac{1}{2}\left(\frac{3}{2} - |s|\right)^2 & \text{for } \frac{1}{2} \leq |s| < \frac{3}{2} \\ 0 & \text{otherwise} \end{cases} \tag{8}$$

- Piecewise Cubic Spline (PCS)

$$W^{(4)}(s) = \begin{cases} \frac{1}{6}\left(4 - 6\,s^2 + 3\,|s|^3\right) & \text{for } 0 \leq |s| < 1 \\ \frac{1}{6}\left(2 - |s|\right)^3 & \text{for } 1 \leq |s| < 2 \\ 0 & \text{otherwise} \end{cases} \tag{9}$$

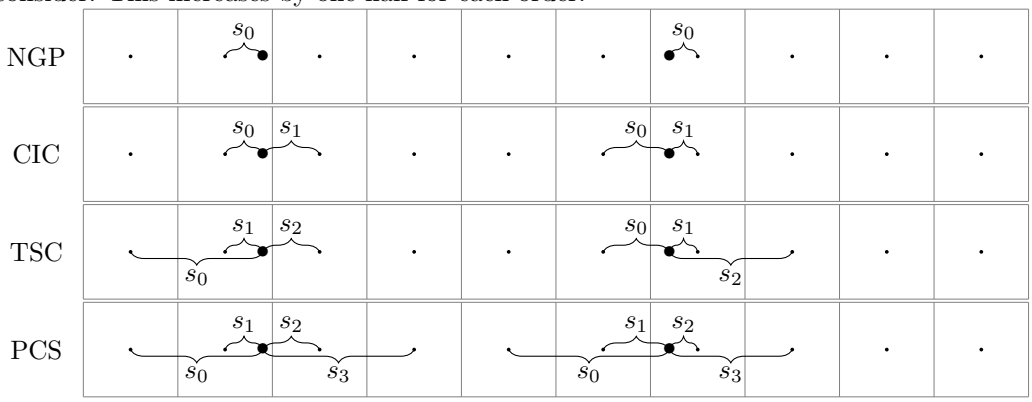


Equations 6 through 9 describe the weight (how much of the mass) that each cell should receive. The variable s is the distance from the cell center (or grid point) in units of the grid size. If a particle is at the center of the cell, then s would be zero. If the particle is at the edge of the cell then s would be 0.5. A particle if a different cell would have $s > 0.5$.

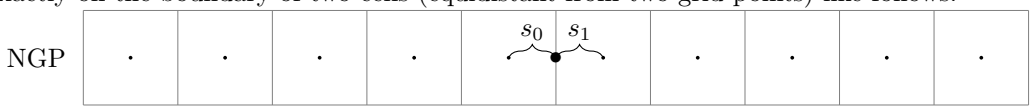
For nearest grid point (NGP) what equation 6 is saying is that if the distance from the particle to the cell center is less than half a grid spacing (i.e., it is in the cell), then all of the mass should be deposited in this cell. Particle that are in different cells (i.e., $s > 0.5$) get 0 (no) mass. There is a special case here – if a particle is exactly on the boundary between two cells ($s = 0.5$) then it must be assigned to one of the two bordering cells and **not both or neither**. This is exactly what we did previously when we first implemented our NGP mass assignment.

The nearest grid point method is rarely (dare we say never) used because it has a terrible property. As particles move the mass is projected onto a single cell, then suddenly completely moves to a new cell. This causes numerical effects and is not physical – mass should flow evenly across the grid. Higher order schemes solve this problem by assigning different fractions of the mass to multiple cells. The next higher order scheme (cloud-in-cell) spreads the mass over two neighbouring cells ($2^3 = 8$ in 3D). Triangular shaped cloud spreads the mass over 3 cells ($3^3 = 27$ in 3D) while piecewise cubic spline spreads the mass over 4 cells ($4^3 = 64$ in 3D). The higher the order, the more computational expensive it becomes. If it is not obvious, the total mass assigned **must** be conserved.

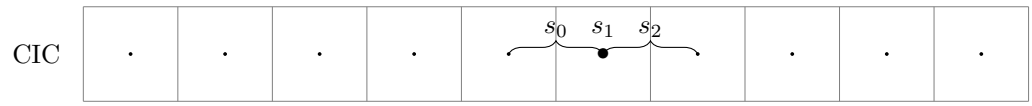
The following diagrams illustrate the different values of s for each of the orders in one dimension. The small dots indicate the center coordinate (grid points) while "•" indicates where two different particles are located. The different s values and their magnitude are marked. For NGP, only grid points that are less than 0.5 away are considered while for CIC grid points that are 1 unit away are consider. This increases by one-half for each order.



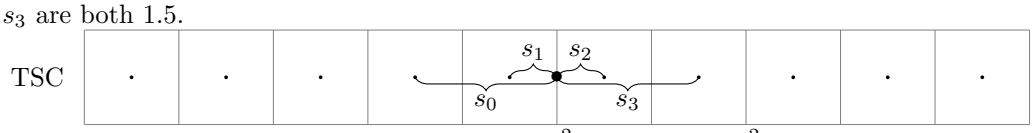
There are special cases for each order that need to be considered. With NGP the particle can be exactly on the boundary of two cells (equidistant from two grid points) like follows.



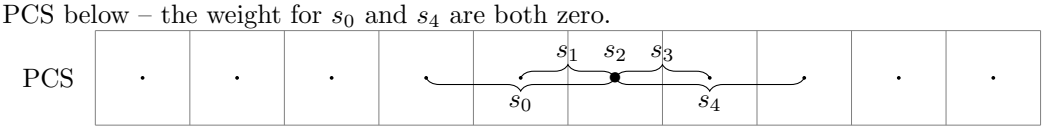
Technically there is no reason to choose one over the other, but it is important to choose only one and to be consistent. In our simple implementation we always choose the one to the right. For CIC we have the case where the point is exactly on the grid point and hence s is zero or exactly one.



Again it doesn't matter which pair you choose, s_0 and s_1 or s_1 and s_2. In fact, the contribution from s_1 is 1 and *both* s_0 and s_2 would contribute zero because their absolute value is one, and $W(s_0) = W(s_2) = 1 - |s| = 1 - |1| = 0$! For TSC the special case is the same, but the distance s_0 and s_3 are both 1.5.



This means that $W(s_0) = W(s_3) = \frac{1}{2}\left(\frac{3}{2} - |s|\right)^2 = \frac{1}{2}\left(\frac{3}{2} - |\frac{3}{2}|\right)^2 = 0$! This means you can choose the three cells s_0, s_1 and s_2 or s_1, s_2 and s_3 – it does not matter. The situation is identical for PCS below – the weight for s_0 and s_4 are both zero.



This special cases also happen to match up with where the different weight functions switch. We can use this to our advantage.

Let's consider PCS. In the following table are the different values of s. You can see that they match the boundary conditions in equation 9. Let's assign the weight equations in two different

ways. You will see that we can use either and this should be obvious because the value of the weigh functions plotted above match when they change between functional forms.

	s	Start with s_0	Start with s_1	Weight
s_0	-2	$\frac{1}{6}\,(2-\lvert s\rvert)^3$		0
s_1	-1	$\frac{1}{6}\left(4-6\,s^2+3\,\lvert s\rvert^3\right)$	$\frac{1}{6}\,(2-\lvert s\rvert)^3$	$\frac{1}{6}$
s_2	0	$\frac{1}{6}\left(4-6\,s^2+3\,\lvert s\rvert^3\right)$	$\frac{1}{6}\left(4-6\,s^2+3\,\lvert s\rvert^3\right)$	$\frac{4}{6}$
s_3	1	$\frac{1}{6}\,(2-\lvert s\rvert)^3$	$\frac{1}{6}\left(4-6\,s^2+3\,\lvert s\rvert^3\right)$	$\frac{1}{6}$
s_4	2		$\frac{1}{6}\,(2-\lvert s\rvert)^3$	0

With some thought you should come to the conclusion that at all orders you should use the "start with s_1" option when exactly at the "special case". Your calculation to select the index of the first cell should choose the s_1 cell. If you move the point the tiniest possible amount ϵ to the left it should then choose the s_0 cell. You can use the index calculation from NGP as a model as that is exactly what it does. Once you have the starting cell index, and number of cells to consider (which increases with order) then you can trivially calculate the s values and apply the functional forms in order, and it should then work in all cases. No need for conditional statements at all!